Head & Neck Anatomy: Gross Structures & Development of the Ear
Master Sequence Followed: Sequence 1

1. The external ear consists of the auricle (pinna) and the external acoustic meatus. The resilient structural framework of the auricle is composed of which type of tissue?
a. Hyaline cartilage
b. Yellow elastic cartilage
c. White fibrocartilage
d. Dense compact bone
e. Spongy trabecular bone
2. The lowest, soft dependent part of the auricle is completely devoid of cartilage and contains only fibrofatty tissue. This specific landmark is the:
a. Helix
b. Tragus
c. Lobule (earlobe)
d. Antihelix
e. Concha
3. The external acoustic meatus is an S-shaped canal that leads from the exterior to the tympanic membrane. What is the approximate total length of this canal in an adult?
a. 12 mm
b. 18 mm
c. 36 mm
d. 24 mm (about 1 inch)
e. 50 mm
4. The external acoustic meatus is structurally heterogeneous. It is divided into an outer cartilaginous part and an inner bony part matching which relative proportions?
a. Outer one-half cartilaginous, inner one-half bony
b. Outer two-thirds cartilaginous, inner one-third bony
c. Outer three-fourths cartilaginous, inner one-fourth bony
d. Outer one-fourth cartilaginous, inner three-fourths bony
e. Outer one-third cartilaginous, inner two-thirds bony
5. Modified sweat glands located within the subcutaneous tissue of the cartilaginous portion of the meatus secrete a protective, waxy substance (cerumen). These are the:
a. Sebaceous glands
b. Ceruminous glands
c. Parotid acini
d. Meibomian glands
e. Lacrimal units
6. To inspect the tympanic membrane through an otoscope in an adult, the clinician must pull the auricle in which direction to straighten the S-shaped curvature of the meatus?
a. Downward and backward
b. Upward, backward, and laterally
c. Straight forward
d. Downward and forward
e. Purely medially

7. The tympanic membrane (eardrum) forms the thin, translucent partition between the external acoustic meatus and the middle ear. It is set obliquely so that its outer surface faces:
a. Directly superiorly
b. Directly medially
c. Downward, forward, and laterally
d. Upward and backward
e. Purely posteriorly
8. The center of the outer surface of the tympanic membrane is drawn inward, producing a concave depression. The deepest central point of this depression corresponds to the attachment of the handle of the malleus and is called the:
a. Pars flaccida
b. Pars tensa
c. Cone of light
d. Umbo
e. Scleral spur
9. When checking a healthy tympanic membrane with an otoscope, a bright triangular reflection of light is seen radiating from the umbo. This landmark is located in which quadrant?
a. Anterosuperior quadrant
b. Anteroinferior quadrant
c. Posterosuperior quadrant
d. Posteroinferior quadrant
e. Central apex zone
10. The vast majority of the tympanic membrane is tense, rigid, and tightly anchored to the surrounding bone by a fibrocartilaginous ring. This main vascularized area is the:
a. Pars flaccida (Shrapnell's membrane)
b. Pars tensa
c. Tegmen tympani
d. Promontory face
e. Fenestra vestibuli
11. The small, thin, triangular upper portion of the tympanic membrane situated above the anterior and posterior malleolar folds is flaccid and lacks a middle fibrous layer. This is the:
a. Pars tensa
b. Umbo dome
c. Pars flaccida
d. Prominence of lateral canal
e. Pyramidal eminence
12. The middle ear (tympanic cavity) is an air-filled space housed entirely within which bone?
a. Sphenoid bone greater wing
b. Squamous temporal bone
c. Mastoid temporal bone
d. Petrous part of the temporal bone
e. Occipital bone basilar part

13. The roof of the middle ear cavity is formed by a thin plate of bone that separates it directly from the middle cranial fossa and the temporal lobe of the brain. This plate is the:
a. Jugular wall
b. Carotid wall
c. Tegmen tympani
d. Promontory
e. Mastoid wall
14. The floor of the middle ear cavity is a thin sheet of bone that isolates it from a major vascular channel situated immediately beneath it. This channel is the:
a. Internal carotid artery
b. Internal jugular vein (superior bulb)
c. Middle meningeal artery
d. Sigmoid sinus
e. Transverse sinus
15. The anterior wall of the middle ear is unique because it features a large opening leading into a osteofibrous channel that connects the middle ear to the nasopharynx. This channel is the:
a. External acoustic meatus
b. Auditory (Eustachian) tube
c. Aditus ad antrum
d. Canalicus chordae tympani
e. Carotid canal exit
16. The auditory tube is essential for hearing because its primary physiological function is to:
a. Secrete endolymph fluid
b. House the auditory ossicles
c. Equalize air pressure on both sides of the tympanic membrane
d. Amplify high-frequency sound waves
e. Dampen vibrations of the stapes
17. The posterior wall of the middle ear features an opening in its upper section that leads directly backward into the mastoid air cells. This opening is the:
a. Fenestra cochleae
b. Foramen rotundum
c. Aditus ad antrum
d. Internal acoustic meatus
e. Petrosquamous fissure
18. The medial wall of the middle ear is formed by the lateral wall of the internal ear. It exhibits a rounded projection caused by the basal turn of the cochlea, which is named the:
a. Pyramidal eminence
b. Promontory
c. Fossula of the fenestra
d. Tegmen ridge
e. Cochleariform process
19. Located superior and posterior to the promontory on the medial wall is an oval opening that is completely closed in life by the footplate of the stapes. This is the:
a. Round window (Fenestra cochleae)

b. Oval window (Fenestra vestibuli)
c. Foramen spinosum
d. Foramen lacerum
e. Hiatus of facial canal

20. Situated inferior and posterior to the promontory is a round opening closed by the secondary tympanic membrane. This structure is the:

a. Fenestra vestibuli
b. Fenestra cochleae (Round window)
c. Foramen ovale
d. Aditus entry
e. Cochlear duct

21. The middle ear contains a chain of three tiny, articulating bones called the auditory ossicles. Moving from the tympanic membrane to the oval window, what is their correct sequence?

a. Stapes, Incus, Malleus
b. Malleus, Stapes, Incus
c. Incus, Malleus, Stapes
d. Malleus, Incus, Stapes
e. Incus, Stapes, Malleus

22. The handle of the malleus is structurally attached to the inner surface of the tympanic membrane, while its rounded head articulates directly with the:

a. Stapes footplate
b. Promontory bone
c. Body of the incus
d. Lenticular process
e. Pyramidal eminence

23. The stapes is the smallest bone in the human body. Its footplate (base) fits tightly inside the margin of the oval window, where it transmits mechanical sound vibrations to the:

a. Air cells of the mastoid
b. Perilymph fluid of the internal ear
c. Mucosal layer of the auditory tube
d. External acoustic meatus
e. Subarachnoid space

24. The tensor tympani muscle works to pull the handle of the malleus medially, tensing the tympanic membrane to protect the ear from loud sounds. It is innervated by which nerve?

a. Facial nerve (CN VII)
b. Glossopharyngeal nerve (CN IX)
c. Mandibular nerve (V3 branch of Trigeminal)
d. Chorda tympani nerve
e. Vagus nerve (CN X)

25. The stapedius muscle is embedded within the pyramidal eminence and attaches to the neck of the stapes to dampen its movements. It is innervated by a branch of which nerve?

a. Trigeminal nerve (V3)
b. Facial nerve (CN VII)
c. Glossopharyngeal nerve (CN IX)

d. Vestibulocochlear nerve (CN VIII)
e. Accessory nerve (CN XI)

26. The chorda tympani nerve passes directly through the middle ear cavity by running across the medial surface of the tympanic membrane, passing between the:
a. Incus and stapes
b. Malleus and incus
c. Handle and head of malleus purely
d. Promontory and round window
e. Tegmen and floor plate

27. The internal ear, or labyrinth, is located inside the petrous temporal bone. It consists of a bony labyrinth that completely houses a delicate fluid-filled:
a. Auditory canal
b. Membranous labyrinth
c. Mastoid antrum
d. Tympanic cavity
e. Middle ear recess

28. The space between the outer walls of the bony labyrinth and the inner walls of the membranous labyrinth is filled with a specialized fluid called:
a. Endolymph
b. Perilymph
c. Cerebrospinal fluid
d. Lymphatic fluid
e. Vitreous fluid

29. The interior of the membranous labyrinth itself is filled with a separate, unique fluid high in potassium ions, known as:
a. Perilymph
b. Aqueous humor
c. Endolymph
d. Plasma fluid
e. Synovial fluid

30. The bony labyrinth consists of three parts: the vestibule, the semicircular canals, and a snail-shaped shell that holds the organ of hearing. This snail-shaped shell is the:
a. Utricle
b. Saccule
c. Cochlea
d. Ampulla
e. Infundibulum

31. The membranous part of the cochlea contains the receptor organ for hearing, which features specialized sensory hair cells sitting on the basilar membrane. This organ is the:
a. Macula
b. Crista ampullaris
c. Spiral organ (Organ of Corti)
d. Otolithic membrane
e. Saccular cupula

32. The membranous semicircular ducts contain specialized sensory receptors located within their expanded ends (ampullae) that detect dynamic rotational acceleration of the head. These are the:
a. Organs of Corti
b. Cristae ampullares
c. Maculae
d. Otolith organs
e. Spiral ganglia
33. The utricle and saccule are two membranous sacs located within the bony vestibule. They contain sensory patches that monitor linear acceleration and gravity, known as the:
a. Cristae
b. Maculae
c. Basilar tracks
d. Helicotrema paths
e. Tectorial sheets
34. A 4-year-old child presents with a severe upper respiratory tract infection that tracks upward through the short, horizontal auditory tube. This results in an infection of the middle ear cavity called:
a. Otitis externa
b. Otitis interna (labyrinthitis)
c. Otitis media
d. Mastoiditis
e. Myringitis
35. If acute otitis media is left untreated, pus can accumulate and exert high pressure inside the cavity. This can lead to an acute mechanical complication characterized by:
a. Fusion of the semicircular canals
b. Perforation of the tympanic membrane
c. Selective atrophy of the pinna
d. Elongation of the styloid process
e. Ossification of the external meatus
36. Surgical incision of the tympanic membrane to drain pus in cases of severe otitis media is called a myringotomy. To avoid damaging the chorda tympani nerve and the auditory ossicles, the incision is safely performed in which quadrant?
a. Posterosuperior quadrant
b. Anteroforward tip apex
c. Posteroinferior (or anteroinferior) quadrant
d. Central umbo dome directly
e. Superior-most pars flaccida zone
37. Chronic middle ear infections can erode through the thin bony roof (tegmen tympani). This allows bacteria to track directly into the cranial cavity to cause:
a. Submandibular sialadenitis
b. Cavernous sinus syndrome
c. Meningitis or a temporal lobe brain abscess
d. Bell's palsy on the opposite side
e. Hydrocephalus of the fourth ventricle

38. Paralysis of the stapedius muscle due to a lesion of the facial nerve proxmial to the pyramidal eminence causes a clinical condition where normal sounds are perceived as intolerably loud. This is:
a. Presbycusis
b. Tinnitus
c. Vertigo
d. Hyperacusis
e. Anacusis
39. Developmentally, the very first indication of the ear appears in the early fourth embryo week as a pair of ectodermal thickenings on either side of the hindbrain called:
a. Otic pits
b. Otic placodes
c. Lens placodes
d. Branchial grooves
e. Pharyngeal pouches
40. The otic placodes soon invaginate below the surface ectoderm and pinch off completely to form a hollow, fluid-filled epithelial vesicle known as the:
a. Optic cup
b. Otic vesicle (otocyst)
c. Lens vesicle
d. Cervical sinus
e. Stomodeum capsule
41. The otic vesicle is a vital embryonic primordium that differentiates to give rise to the entire structure of the:
a. External ear canal
b. Middle ear ossicles chain
c. Membranous labyrinth
d. Bony labyrinth capsule
e. Tympanic membrane external coat
42. As the otic vesicle elongates, it divides into a dorsal utricular component and a ventral saccular component. The dorsal utricular component sprouts three flat discs that give rise to the:
a. Auditory ossicles
b. Cochlear duct channels
c. Semicircular ducts
d. Mastoid air spaces
e. Eustachian tubes
43. The ventral saccular component of the otic vesicle grows a tubular, spiraling outpocketing that loops two and three-quarter times to form the definitive:
a. Semicircular canal
b. Cochlear duct (scala media)
c. Utricle sac
d. Endolymphatic duct
e. Tympanic antrum
44. The mesenchyme surrounding the developing membranous labyrinth undergoes chondrification and later vacuolizes to form spaces filled with perilymph. This surrounding shell ossifies to become the:

a. Auditory ossicles chain
b. Temporal squamous plate
c. Bony labyrinth
d. Styloid process line
e. Mastoid cortex

45. The tympanic cavity (middle ear) and the auditory tube develop entirely from the internal endodermal outpocketing of the:

a. First pharyngeal cleft
b. Second pharyngeal pouch
c. First pharyngeal pouch (tubotympanic recess)
d. Third pharyngeal pouch
e. Cervical sinus of His

46. The malleus and incus ossicles develop from the mesenchymal condensation of the cartilage scaffold of the:

a. First pharyngeal arch (Meckel's cartilage)
b. Second pharyngeal arch (Reichert's cartilage)
c. Third pharyngeal arch
d. Fourth pharyngeal pouch
e. Otic vesicle wall

47. The stapes ossicle develops from the dorsal mesenchyme segment belonging to the cartilage scaffold of the:

a. First pharyngeal arch
b. Second pharyngeal arch (Reichert's cartilage)
c. Third pharyngeal arch
d. Sixth pharyngeal arch
e. Hindbrain neural crest rows

48. The external acoustic meatus develops from the deep invagination of the external ectodermal groove known as the:

a. First pharyngeal pouch
b. First pharyngeal cleft (groove)
c. Second pharyngeal cleft
d. Cervical sinus track
e. Stomodeal pit

49. The tympanic membrane develops from three separate germ layers, mirroring its function as a boundary. Its internal epithelial lining is derived from the endoderm of the:

a. First pharyngeal cleft
b. First pharyngeal pouch (tubotympanic recess)
c. Surface ectoderm sheet
d. Neural crest core
e. Preotic somite rows

50. The six small elevations of mesenchyme that appear around the margins of the first pharyngeal cleft and fuse together to form the definitive external auricle (pinna) are the:

a. Otic placodes
b. Palatal shelves
c. Auricular hillocks (of His)
d. Branchial arches

e. Lingual swellings

Answer Key Table (Ear Gross Anatomy & Development - Sequence 1)

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-48	Q 49-50

b	**b**	**c**	**b**	**b**	**b**	**b**
c	**b**	**b**	**b**	**c**	**c**	**c**
d	**c**	**b**	**b**	**c**	**b**	
e	**b**	**c**	**b**	**d**	**c**	
b	**b**	**b**	**a**	**b**	**c**	
b	**c**	**b**	**d**	**c**	**a**	
c	**c**	**a**	**c**	**c**	**b**	
d	**b**	**c**	**b**	**b**	**b**	